

Automated Indian currency note damage detection and severity assessment using image processing and deep learning.

Background:

over the time, currency notes undergo physical deterioration due to continuous circulation, mishandling, environmental exposure and accidental damage. Common defects include tears, folds, stains, missing corners, wrinkles, tape repairs, ink marks, water damage.

Banks and financial institutions currently rely on manual inspection to identify damaged notes. This process is time-consuming, subjective, and prone to inconsistencies. An automated image processing system can significantly improve the speed, consistency, and accuracy of currency note inspection.

This project proposes an intelligent image processing system capable of analyzing images of Indian currency notes, detecting visible physical damage, classifying the type of damage, and estimating the severity of damage using computer vision and deep learning techniques.

Problem statement:

Manual identification of damaged currency notes is inefficient and subjective. Existing systems primarily focus on recognising the denomination or detecting counterfeit notes, while limited work has been done on automated damage assessment of Indian currency notes.

The objective of this project is to develop an end-to-end image processing system that automatically detects damage in Indian currency notes and classifies the damage into different categories along with estimating its severity.

Objectives:

The proposed system aims to:

- Acquire images of Indian currency notes.
- Perform image preprocessing to improve image quality.
- Segment the currency note from the background.
- Detect damaged regions.
- Classify the type of damage.
- Estimate damage severity.
- Evaluate the model using standard image processing metrics.

Scope of the project:

The proposed system focuses only on physical damage detection.

Damage categories include:

- Tear
- Fold
- Wrinkle
- Missing Corner
- Dirt/Stain
- Tape Repair
- Ink Mark

About the dataset:

Indian Currency Note Images Dataset 2020

Source: <https://www.kaggle.com/datasets/vishalmane109/indian-currency-note-images-dataset-2020>

The dataset contains high-quality images of Indian currency notes captured under different backgrounds and lighting conditions.

Denominations include

- ₹10
- ₹20
- ₹50
- ₹100
- ₹200
- ₹500

- ₹2000

The dataset consists primarily of clean currency note images.

Dataset enhancement:

Since the dataset does not contain damaged notes, additional damaged images will be generated using two approaches:

Synthetic Damage Generation

Artificial damage will be introduced using OpenCV.

Types include:

- Fold
- Tear
- Wrinkle
- Shadow
- Dirt
- Blur
- Tape

Manual Image Collection

A small number of damaged notes (or printed replicas) will be photographed under controlled lighting to improve dataset diversity.

Expected dataset size: approximately 4000-6000 images.

Stages in the project:

Step1: Image Acquisition

Input image of an Indian currency note.

Step 2: Image Preprocessing

Operations

- Resize
- RGB normalization
- Noise removal
- Contrast enhancement
- Histogram Equalization
- CLAHE

Purpose

Improve image quality before analysis.

Step 3: Image Enhancement

Techniques

- Gaussian Blur
- Median Filtering
- Bilateral Filtering

Purpose

Reduce unwanted noise while preserving edges.

Step 4: Background Removal

Techniques

- Thresholding
- Contour Detection

Purpose

Extract only the currency note.

Step 5: Image Segmentation

Methods

- Otsu Thresholding
- Adaptive Thresholding
- Morphological Operations

Purpose

Identify damaged regions.

Step 6: Edge Detection

Methods

- Canny Edge Detection
- Sobel Operator

Purpose

Locate tears and missing edges.

Step 7: Feature Extraction

Color Features

Texture Features

Shape Features

Contour Area

Perimeter

GLCM Features

LBP Features

Edge Density

Classification

Two approaches will be compared.

Traditional Machine Learning

Support Vector Machine

Random Forest

Decision Tree

Step 8: Damage Severity Estimation

The damaged region will be segmented.

Damage Percentage

Step 9: output result

Software requirements:

Programming Language

Python

Libraries

OpenCV

NumPy

Pandas

Matplotlib

Scikit-learn

TensorFlow

Keras

Scikit-image

Expected output of the project:

₹500 Currency Note

↓

Detected Damage Small Tear

↓

Severity 12%

↓

Classification Moderately Damaged

↓

Confidence 97.4%

Applications:

Banks

ATMs

Cash Sorting Machines

Currency Exchange Centers

Retail Stores
Reserve Bank of India
Financial Institutions
Educational Research